

The empirical recurrence for OEIS A251316, recovered from its tail

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Abstract

OEIS A251316 records “Empirical recurrence of order 68” and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 256$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 3$, so the recurrence is proved for every $n \geq 71$. Its last coefficient is $c_{68} = -1 \neq 0$, so no further tail satisfies anything shorter; any order-68 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry’s recurrence is the one displayed here, whatever its linked file contains, and it is proved.

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1 A conjecture that is not written down

OEIS A251316 is “Number of $(n+1) \times (7+1)$ 0..1 arrays with no 2×2 subblock having the maximum of its diagonal elements greater than the absolute difference of its antidiagonal elements.”. It has offset 1 and begins

2189, 20484, 186162, 1720147, 16024915, 149982413, 1401489913, 13088358653, ...

The entry states:

Empirical recurrence of order 68 (see link above)

As of the “Last modified” line on the live entry (Jul 23 2025 13:16:18, revision 6) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry’s condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u , and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) = \iota^\top M^{j(n)} \tau$$

for a linear reindexing j fixed by the entry’s shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 256$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 256$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \dots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S' + 1$ consecutive values of a . $\square$

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 68: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 3$ is the first whose minimal order is exactly the stated 68.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 512$ exact terms from $n = 3$ on, it returns order 68 — the order the entry states — with integer coefficients:

$$a(n) = \sum_{i=1}^{68} c_i a(n-i),$$

c_1	55	c_{18}	2345914154391	c_{35}	−82945659049826	c_{52}	−21523856774
c_2	−1484	c_{19}	−36393244272833	c_{36}	133357516999083	c_{53}	6464679339
c_3	26436	c_{20}	90793197110565	c_{37}	−142304018148496	c_{54}	−754241109
c_4	−350306	c_{21}	−130258862788934	c_{38}	98528405139757	c_{55}	−424177285
c_5	3669575	c_{22}	68078094561719	c_{39}	−25084135140443	c_{56}	353742788
c_6	−31470216	c_{23}	220573310606298	c_{40}	−31347931335171	c_{57}	−136666853
c_7	225856057	c_{24}	−835072847225614	c_{41}	45790618297342	c_{58}	30945793
c_8	−1376259499	c_{25}	1755037814704498	c_{42}	−30145667873293	c_{59}	−5084963
c_9	7188941360	c_{26}	−2781677996473068	c_{43}	9754319501408	c_{60}	688361
c_{10}	−32380787866	c_{27}	3584783829535471	c_{44}	1251919530524	c_{61}	−18361
c_{11}	126105980501	c_{28}	−3857806964849776	c_{45}	−3248178260506	c_{62}	−19148
c_{12}	−424338702825	c_{29}	3490435790043326	c_{46}	1773299359901	c_{63}	−65
c_{13}	1227510310292	c_{30}	−2637355061734583	c_{47}	−420451860028	c_{64}	29
c_{14}	−3016165525429	c_{31}	1629911678881276	c_{48}	−79424162249	c_{65}	183
c_{15}	6134778129742	c_{32}	−789089601860925	c_{49}	132373209214	c_{66}	−24
c_{16}	−9713731730378	c_{33}	266885913092610	c_{50}	−87658922174	c_{67}	0
c_{17}	9733602408261	c_{34}	−20189915591987	c_{51}	48210211166	c_{68}	−1

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 71$, and it is the recurrence OEIS A251316 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 3$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{68} - c_1 x^{67} - \dots - c_{68}$. Its constant term is $-c_{68} = 1$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k \lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$

vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 68 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 68, so $\deg q = 68$, and it is monic. It holds from some index t on. From $\max(t, 3)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 68, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 16 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 68, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry’s.

Third, the recovered recurrence was evaluated directly on the entry’s own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is -1 .

Remark 4. The conjecture’s own statement was never read. The entry pins it down to the family of order-68 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

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